Easy Introduction to Neural Networks

Prof. Marios S. Pattichis, FEAMBES

Image and Video Processing and Communications Lab (ivPCL)
Dept. of ECE, University of New Mexico, USA
www.ivpcl.org

Outline

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Background check

What do you already know about the following:

- Plotting 2D points?
- Linear equations: $y = 2x + 1$?
- Inequalities: $y \geq x$?

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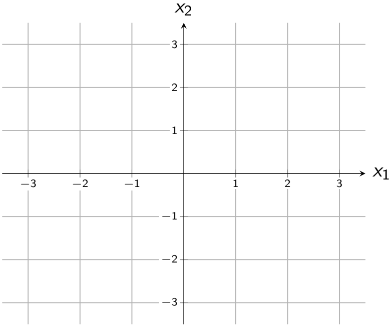
Plot Points



Plot the following points:

- (0, 0).
- (1, 2).
- (3, 4).
- (-1, 2).
- (-3, -1).

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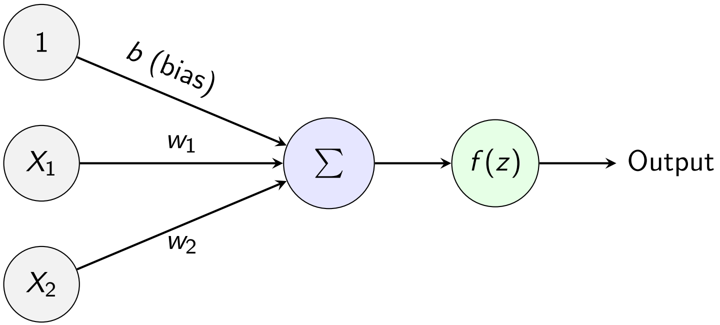
Neural Network Architecture: Single Neuron



$$z = w_1 X_1 + w_2 X_2 + b$$

A 2-input neuron defines a linear equation

Suppose that we want to determine when z is positive:

$$w_1 X_1 + w_2 X_2 + b \geq 0$$

Subtract $w_2 X_2$ from both sides to get:

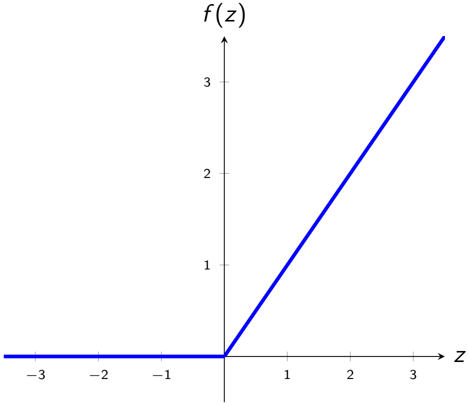
$$w_1 X_1 + \cancel{w_2 X_2} + b - \cancel{w_2 X_2} \geq -w_2 X_2$$

When $w_2 > 0$, dividing by $-w_2$ reverses the inequality:

$$\frac{-w_1}{w_2} X_1 - \frac{b}{w_2} \leq X_2$$

Thus, we can study outputs from a single neuron using inequalities!

Afterwards, apply Activation Function: ReLU



$$f(z) = \text{ReLU}(z) = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{if } z \geq 0 \end{cases}$$

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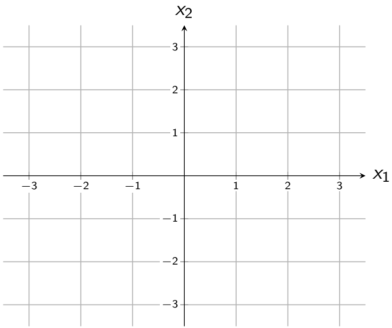
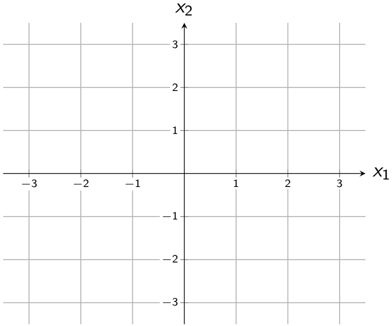
Plotting Inequalities (I of V)



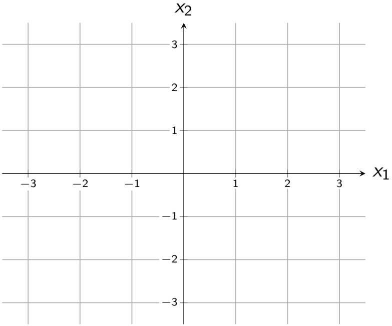
Plot $x_1 \geq 0$**.**

Plotting Inequalities (II of V)



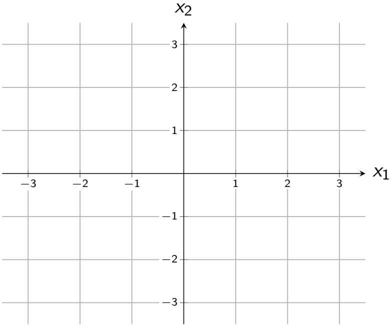
Plot $x_2 \geq 0$.

Plotting Inequalities (III of V)



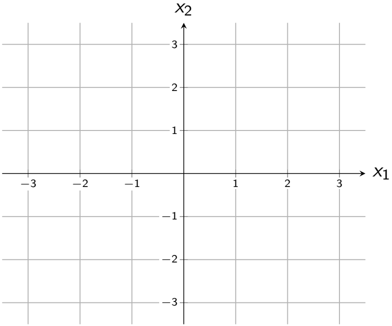
Plot $x_2 \geq x_1$**.**

Plotting Inequalities (IV of V)



Plot $x_2 \geq x_1 + 1$.

Plotting Inequalities (V of V)



Plot $x_2 \geq 2 - x_1$**.**

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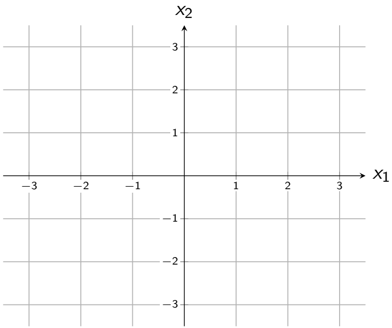
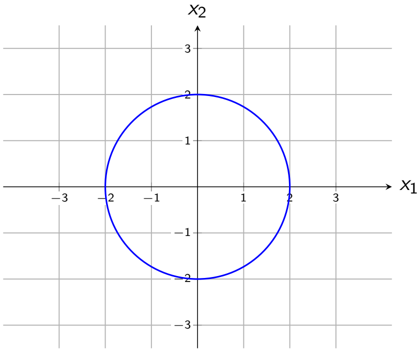
Plotting Multiple Inequalities



Plot the following:

- $x_1 \geq 0$.
- $x_2 \geq 0$.
- $x_1 \leq 1$.
- $x_2 \leq 1$.

Approximating a circular region



Sketch lines around the circle so that the region where all the inequalities are satisfied is **inside the circle**.

Outline

Classification

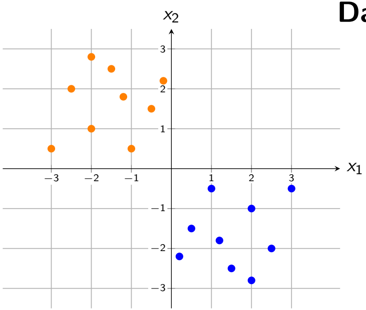
Consider a covid test that relies on two measurements to determine whether one has covid.

We measure two numbers (X_1, X_2) from each patient.

We want to plot the measurements as points as follows:

Orange points: Patient has Covid.
Blue points: Patient is normal.

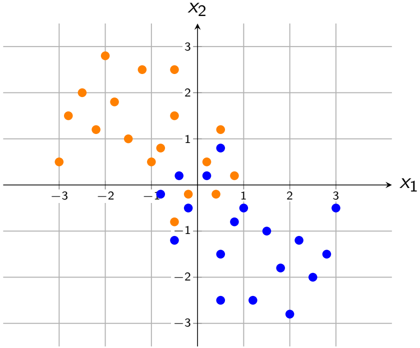
Classification: Linearly Separable Data Categories



Dataset Classification:

- **Orange:** Covid patients
- **Blue:** Normal patients
- Draw a line that separates the two categories.
- Is there a line that is better than another? Discuss within your group.

A more realistic case



Dataset Classification:

- **Covid**
- **Normal**
- Can we draw a line that separates the two categories? Discuss.
- Is there a line that is better than another? Discuss within your group.

Outline

Neural Net is a function with an error (or loss)

Assign the following values for the neural network to learn:

$+1$ for Covid.
-1 for Normal.

We then **want** to train a Neural network so that:

$f(x_1, x_2) = +1$ when (x_1, x_2) is for Covid.
$f(x_1, x_2) = -1$ when (x_1, x_2) is Normal.

When the neural network gets it wrong, it calculates its error (or loss). A simple error is the **squared error**:

$(f(x_1, x_2) - 1)^2$ when (x_1, x_2) is for Covid.
$(f(x_1, x_2) + 1)^2$ when (x_1, x_2) is Normal.

What is the loss when the neural network gets everything right?

Training versus test loss

We often want to build a **balanced dataset** with an equal number of members from each category.

Thus, for 100 patients, how many will be Covid cases and how many will be normal?

Furthermore, to measure how well the neural network will perform on new patients, we randomly select a percentage of the patients (e.g., 20%) and save them for **testing** the network **after training**.

A simple setup

Suppose that we are given 100 points for 50 normals and 50 Covid cases.

We randomly select 20 of them (10 normals + 10 Covid cases) to remove from the dataset for training the neural network. After training, we use these 20 cases to measure the **testing loss**.

We use the 80 remaining ones (40 normals + 40 Covid cases) for training. During training, we use these 80 cases, to measure the **training loss**.

Assigning the training

Very good fit: Both the training and test losses are low.

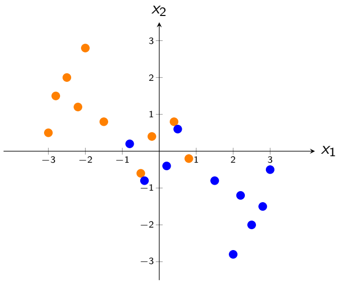
Underfitting: Training and testing losses are high.

Overfitting: Training loss is low and testing loss is high.

Sample Training and Testing Sets

Easy Intro to NN

M.S. Pattichis



- Select 8 Covid and 8 normal points for training.
- Select 2 Covid and 2 normal points for testing.
- Is your testing set fair? Explain.
- What would be an unfair testing set?
- Why is it important to have a good test set? What could go wrong if the test set is bad?